

Deep Feedforward Neural Networks

Nonlinear Representation Learning and Backpropagation

Yasser Hifny

Faculty of Computers and Artificial Intelligence
Helwan University

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Motivation

Deep feedforward neural networks consist of multiple layers of neurons that can learn complex patterns and representations from input data.

If every layer uses a linear activation, stacking layers does not increase the expressive power:

$$y = W^L W^{L-1} \dots W^1 x = Wx.$$

Therefore, the complete network is still only a linear transformation of the input.

Key idea

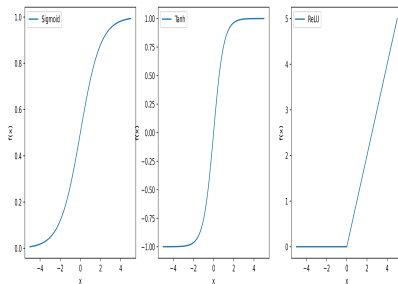
Nonlinear activation functions are required to make a deep network capable of representing nonlinear relationships.

Nonlinear Activation Functions

We introduce nonlinear functions such as:

- sigmoid,
- tanh,
- ReLU.

They can map inputs into different ranges and allow the network to form nonlinear decision boundaries.



Three activation functions commonly used in deep networks.

Deep Network with Nonlinear Activations

With nonlinear activations, a deep network can be written as

$$y = W^L g(W^{L-1} \cdots g(W^2 g(W^1 x + b^1) + b^2) \cdots + b^{L-1}) + b^L.$$

- Each layer performs a linear transformation followed by a nonlinear transformation.
- The composition of these transformations creates expressive representations.
- Nonlinear networks can approximate continuous functions.

Why XOR Requires Nonlinearity

The XOR truth table is

x_1	x_2	t
0	0	0
0	1	1
1	0	1
1	1	0

A linear decision boundary would require

$$x_1 w_1 + x_2 w_2 \leq \theta.$$

For (1, 0) and (0, 1) to be positive:

$$w_1 > \theta, \quad w_2 > \theta.$$

But (1, 1) must be negative:

$$w_1 + w_2 < \theta,$$

which contradicts the first two inequalities.

XOR: Consequence

There is no single linear decision boundary that separates the four XOR points into the required two classes.

Therefore, the XOR problem requires a nonlinear neural network with at least one hidden layer.

Linear layers alone $\nRightarrow$ nonlinear decision boundaries

Hidden layer + nonlinear activation $\Rightarrow$ nonlinear representation

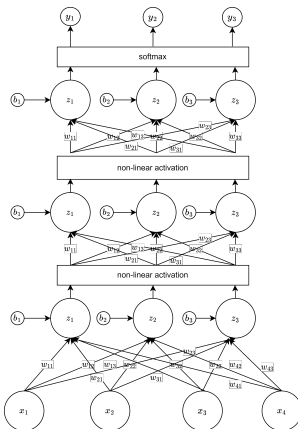
Output Layer and Learning Task

The output layer is selected according to the prediction task.

Task	Output activation	Loss function
Regression	Linear	Mean squared error
Binary classification	Sigmoid	Binary cross-entropy
Multiclass classification	Softmax	Categorical cross-entropy

Our main DNN example uses a softmax output for a multiclass classification problem.

Deep Feedforward Neural Network



Two hidden layers, nonlinear hidden activations, and a three-node softmax output layer.

Network Model

Consider a network with two hidden layers and an output layer.

$$z^{[l]} = W^{[l]}a^{[l-1]} + b^{[l]}, \quad a^{[l]} = \text{sigmoid}(z^{[l]})$$

for the hidden layers, with

$$a^{[3]} = \text{softmax}(z^{[3]}).$$

- x : input vector.
- $W^{[1]}, W^{[2]}, W^{[3]}$: weight matrices.
- $b^{[1]}, b^{[2]}, b^{[3]}$: bias vectors.
- $a^{[0]} = x$.

Forward Propagation

Hidden layer 1

$$z^{[1]} = W^{[1]}x + b^{[1]}, \quad a^{[1]} = \text{sigmoid}(z^{[1]}).$$

Hidden layer 2

$$z^{[2]} = W^{[2]}a^{[1]} + b^{[2]}, \quad a^{[2]} = \text{sigmoid}(z^{[2]}).$$

Output layer

$$z^{[3]} = W^{[3]}a^{[2]} + b^{[3]}, \quad a^{[3]} = \text{softmax}(z^{[3]}).$$

Learning via Backpropagation

Backpropagation trains the network by combining a forward pass with a backward pass.

- ➊ Propagate the input forward through all layers.
- ➋ Compute the prediction error using the target and output.
- ➌ Propagate the error backward using the chain rule.
- ➍ Compute gradients for weights and biases.
- ➎ Update the parameters using gradient descent.

Forward pass → Loss → Backward pass → Update

Softmax Output

For K classes, the softmax output is

$$a_i^{[3]} = \frac{e^{z_i^{[3]}}}{\sum_{j=1}^K e^{z_j^{[3]}}}.$$

The outputs form a probability distribution:

$$a_i^{[3]} \geq 0, \quad \sum_{i=1}^K a_i^{[3]} = 1.$$

For a one-hot target t , the cross-entropy loss is

$$E = - \sum_{i=1}^K t_i \log a_i^{[3]}.$$

Softmax–Cross-Entropy Gradient

We derive the gradient using the chain rule:

$$\frac{\partial E}{\partial z} = \frac{\partial a}{\partial z} \frac{\partial E}{\partial a}.$$

For the correct class j ,

$$\frac{\partial E}{\partial a} = \begin{pmatrix} 0 \\ \vdots \\ -1/a_j \\ \vdots \\ 0 \end{pmatrix}.$$

The softmax Jacobian is

$$\frac{\partial a_i}{\partial z_j} = a_i(\delta_{ij} - a_j).$$

Consequently,

$$\boxed{\frac{\partial E}{\partial z} = a - t.}$$

Output-Layer Backpropagation

Define

$$\delta^{[3]} = \frac{\partial E}{\partial z^{[3]}}.$$

For softmax with cross-entropy:

$$\delta^{[3]} = a^{[3]} - t.$$

The parameter gradients are

$$\frac{\partial E}{\partial W^{[3]}} = \delta^{[3]} (a^{[2]})^T$$

$$\frac{\partial E}{\partial b^{[3]}} = \delta^{[3]}$$

Backpropagation Through Hidden Layer 2

Using the chain rule,

$$\delta^{[2]} = (W^{[3]})^T \delta^{[3]} \circ g'(z^{[2]}).$$

For the sigmoid,

$$g'(z) = g(z)(1 - g(z)) = a(1 - a).$$

Therefore,

$$\boxed{\delta^{[2]} = (W^{[3]})^T \delta^{[3]} \circ a^{[2]} \circ (1 - a^{[2]})}$$

and

$$\frac{\partial E}{\partial W^{[2]}} = \delta^{[2]} (a^{[1]})^T, \quad \frac{\partial E}{\partial b^{[2]}} = \delta^{[2]}.$$

Backpropagation Through Hidden Layer 1

Similarly,

$$\delta^{[1]} = (W^{[2]})^T \delta^{[2]} \circ a^{[1]} \circ (1 - a^{[1]})$$

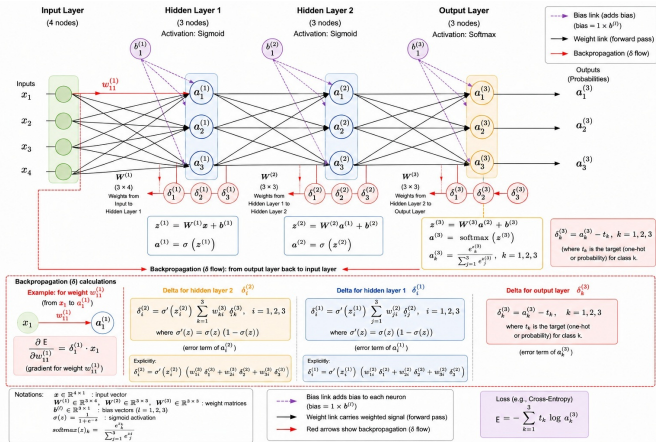
The gradients are

$$\frac{\partial E}{\partial W^{[1]}} = \delta^{[1]} \mathbf{x}^T, \quad \frac{\partial E}{\partial b^{[1]}} = \delta^{[1]}.$$

Notation

$\circ$ denotes element-wise (Hadamard) multiplication.

Backpropagation Overview



Forward propagation computes activations; the backward pass propagates error signals and computes gradients.

Gradient-Descent Updates

For a minibatch or set of N training examples,

$$W^{[l]} \leftarrow W^{[l]} - \eta \frac{1}{N} \sum_{n=1}^N \frac{\partial E^n}{\partial W^{[l]}}$$

$$b^{[l]} \leftarrow b^{[l]} - \eta \frac{1}{N} \sum_{n=1}^N \frac{\partial E^n}{\partial b^{[l]}}.$$

where η is the learning rate.

The forward/backward/update cycle is repeated over multiple epochs.

The Delta Term

The delta term is the derivative of the loss with respect to the pre-activation:

$$\delta^{[l]} = \frac{\partial E}{\partial z^{[l]}}$$

For the output layer:

$$\delta^{[3]} = a^{[3]} - t.$$

For hidden layers, the delta is obtained from the delta of the next layer, the connecting weights, and the derivative of the activation function.

Backpropagation as Repeated Chain Rules

For hidden layer 2:

$$\frac{\partial E}{\partial z^{[2]}} = \frac{\partial E}{\partial z^{[3]}} \frac{\partial z^{[3]}}{\partial a^{[2]}} \frac{\partial a^{[2]}}{\partial z^{[2]}}.$$

Since

$$z^{[3]} = W^{[3]} a^{[2]} + b^{[3]},$$

we obtain

$$\delta^{[2]} = (W^{[3]})^T \delta^{[3]} \circ g'(z^{[2]}).$$

The same pattern is applied recursively to earlier layers.

XOR Example

We provide a NumPy implementation to learn XOR using a two-layer nonlinear network.

$$X = \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}, \quad Y = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 0 \end{bmatrix}.$$

Architecture:

- 2 input variables,
- 4 hidden units,
- 1 sigmoid output,
- random initialization,
- learning rate 0.1,
- 10,000 iterations.

XOR Example: Forward and Backward Pass

```
for epoch in range(epochs):  
    # Forward propagation  
    l0 = X  
    l1 = sigmoid(np.dot(l0, w0) + b0)  
    l2 = sigmoid(np.dot(l1, w1) + b1)  
  
    # Backward propagation  
    l2_error = Y - l2  
    l2_delta = l2_error * sigmoid_derivative(l2)  
  
    l1_error = l2_delta.dot(w1.T)  
    l1_delta = l1_error * sigmoid_derivative(l1)  
  
    # Update parameters  
    w1 += l1.T.dot(l2_delta) * lr  
    b1 += np.sum(l2_delta, axis=0, keepdims=True) * lr  
    w0 += l0.T.dot(l1_delta) * lr  
    b0 += np.sum(l1_delta, axis=0, keepdims=True) * lr
```

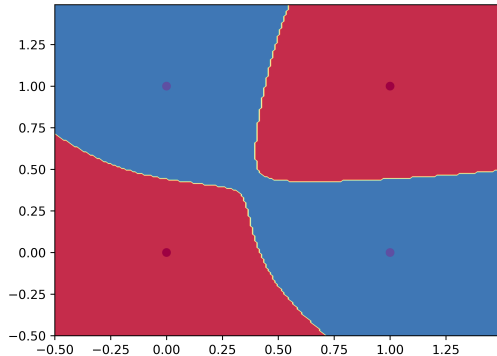
XOR Example: Decision Boundary Code

```
# Plot decision boundary
h = 0.01
x_min, x_max = X[:, 0].min() - 0.5, X[:, 0].max() + 0.5
y_min, y_max = X[:, 1].min() - 0.5, X[:, 1].max() + 0.5
xx, yy = np.meshgrid(
    np.arange(x_min, x_max, h),
    np.arange(y_min, y_max, h))

l0 = np.c_[xx.ravel(), yy.ravel()]
l1 = sigmoid(np.dot(l0, w0) + b0)
l2 = sigmoid(np.dot(l1, w1) + b1)

Z = (l2 > 0.5).astype(int)
Z = Z.reshape(xx.shape)
plt.contourf(xx, yy, Z, cmap=plt.cm.Spectral)
plt.scatter(X[:, 0], X[:, 1], c=Y[:, 0], cmap=plt.cm.Spectral)
plt.savefig('xor_decision_boundary.png', dpi=600)
```

XOR Decision Boundary



The learned nonlinear boundary separates (0, 0) and (1, 1) from (0, 1) and (1, 0).

From Linear to Deep Networks

Linear layers

Composition remains linear.

Nonlinear activations

Intermediate representations
become nonlinear.

Backpropagation

Gradients train the parameters
of all layers.

Deep network = composition of transformations + nonlinearities

Assignment

Implement optical recognition of handwritten digits as a multiclass classification task using a deep feedforward neural network (using Python from scratch, using Keras toolkit, and Google Colab).

Dataset:

<https://archive.ics.uci.edu/ml/datasets/Optical+Recognition+of+Handwritten+Digits>

Expected learning outcome

Apply the DNN model, nonlinear hidden layers, softmax output, categorical cross-entropy, and backpropagation to a real multiclass recognition problem.

Summary

- Stacking linear layers still produces a linear transformation.
- Nonlinear activations enable expressive representations and nonlinear decision boundaries.
- XOR demonstrates why a hidden nonlinear layer is necessary for some problems.
- A DNN can use different output activations and losses according to the task.
- Forward propagation computes the network output.
- Backpropagation applies the chain rule to compute layer-wise gradients.
- The delta terms propagate the error backward through the network.
- Gradient descent updates weights and biases over repeated epochs.

Thank You

Questions?